

W9: COSY Longitudinal Study — 0.5 ps vs 2 ps

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1 Objective

The 11-stage transverse Twiss optimisation produces identical quad currents for both the 0.5 ps and 2 ps operating modes of the UH MkV FEL, as shown analytically in S9 and numerically with FELsim. A reviewer requests COSY INFINITY-level evidence that the full 3D (6D phase space) simulation confirms this, and that the longitudinal beam quality is preserved through the transport line.

This study uses COSY INFINITY’s differential algebra (DA) framework to extract the complete 6D transfer map at the undulator entrance, propagate 10^4 -particle bunches through it for both operating modes, and test whether adding longitudinal FIT objectives changes the optimised solution.

2 Method

2.1 COSY 6D Transfer Map Extraction

The 11-stage COSY FIT optimisation is run with the standard transverse objectives (§3.1) and the following COSY parameters:

- Computation order: 3 (DA polynomials up to 3rd order)
- Dimensions: 3 (6D phase space: x, a, y, b, l, δ where $a = x'$, $b = y'$, $\delta = \delta_K = \Delta K / K_0$)
- Transfer matrix output order: 2 (linear + $(l|\delta\delta)$)
- Fringe field order: 0
- $E = 40 \text{ MeV}$, $\varepsilon_n = 8 \pi \cdot \text{mm} \cdot \text{mrad}$

The full 6×6 linear transfer map $\mathcal{M}$ and second-order coefficient $(l|\delta\delta) = \text{ME}(5, 66)$ are extracted via the COSY PM command.

COSY PM output and $(\delta|\delta)$. COSY’s PM command outputs five map components per line (for x, a, y, b, l), not six: the sixth coordinate δ is the DA independent variable, and its map is the identity by construction in a passive beamline. The six-digit indices (e.g., 000001) confirm that the simulation uses three DA dimensions (6D phase space), not two. The reader code pads the missing sixth component with zero, yielding $(\delta|\delta) = 0$ in the raw output. We set $(\delta|\delta) = 1$ when constructing the full 6×6 matrix, consistent with energy conservation.

Evidence that the simulation is 3D, not 2D:

- Six-digit polynomial indices (`dimensions=2` uses four digits)
- $(l|x) = 9.87 \times 10^{-3}$, $(l|a) = -4.80 \times 10^{-2}$ (non-zero path-length coupling through dipoles)
- $(l|\delta) = 27.09 \text{ mm}$ (non-zero momentum compaction)

2.2 6D Bunch Propagation

Gaussian bunches ($N = 10,000$, seed = 42) are generated in COSY coordinates for four scenarios:

1. 0.5 ps, $h = 0$ (no chirp)
2. 0.5 ps, $h = 5 \times 10^9 \text{ s}^{-1}$
3. 2 ps, $h = 0$
4. 2 ps, $h = 5 \times 10^9 \text{ s}^{-1}$

where h is the linear energy chirp rate: $\delta \rightarrow \delta + h l / (\beta c)$. Each bunch is propagated through the linear map: $\mathbf{X}_{\text{out}} = \mathcal{M} \mathbf{X}_{\text{in}}$, with $(l|\delta\delta)$ correction applied if non-zero.

Beam parameters: $\sigma_x = \sigma_y = 0.8 \text{ mm}$, $\sigma_{x'} = \sigma_{y'} = \varepsilon_{\text{geom}} / \sigma_x$, $\sigma_\delta = 0.5\%$.

2.3 $(l|\delta)$ Objective Test

Two COSY FIT runs are compared:

Baseline Standard 11-stage transverse-only objectives.

Augmented Same objectives plus $(l|\delta) = 0$ at the undulator entrance (element 117), using COSY's ME(5,6) function with unit weight.

3 Results

3.1 Transfer Map Analysis

The transverse optimisation achieves $\text{MSE} = 2.27 \times 10^{-7}$, matching all four Twiss targets to better than 10^{-3} . Table 1 lists the longitudinal map elements at the undulator entrance.

The transfer map has the expected block structure:

- **Transverse block** (4×4): Fully populated, as expected for a beamline with quadrupoles and bends.
- **Path-length row** ($(l|x_j)$): $(l|x)$ and $(l|a)$ are non-zero — off-axis particles travel different path lengths through dipoles. This is a geometric effect of bending magnets, *not* a coupling that affects the independence of transverse matching from bunch length.
- **Energy row** ($(\delta|x_j)$): All zero for $x_j \in \{x, a, y, b\}$ — no energy change from transverse coordinates. This confirms that the transverse and longitudinal subspaces are decoupled in the Twiss matching sense.
- $(l|\delta\delta) = 0$ — no nonlinear bunch compression.

Table 1: Longitudinal elements of the COSY transfer map $\mathcal{M}$ at the undulator entrance ($E = 40$ MeV, FR.0, order 3). COSY notation ($y_i|y_j$) denotes the first-order map coefficient; the transport-code equivalents (R_{ij} , T_{ijk}) are given for reference.

Map element	FOX	Transport	Value	Interpretation
$(l x)$	ME(5,1)	R_{51}	9.874×10^{-3}	Path length from x (dipoles)
$(l a)$	ME(5,2)	R_{52}	-4.799×10^{-2}	Path length from a (dipoles)
$(l y)$	ME(5,3)	R_{53}	0	Path length from y
$(l b)$	ME(5,4)	R_{54}	0	Path length from b
$(l l)$	ME(5,5)	R_{55}	1.000	Path length identity
$(l \delta)$	ME(5,6)	R_{56}	2.709×10^{-2}	Momentum compaction (27.09 mm)
(δx)	ME(6,1)	R_{61}	0	Energy from x
(δa)	ME(6,2)	R_{62}	0	Energy from a
(δy)	ME(6,3)	R_{63}	0	Energy from y
(δb)	ME(6,4)	R_{64}	0	Energy from b
$(\delta \delta)$	ME(6,6)	R_{66}	1.000	Energy identity (implicit)
$(l \delta\delta)$	ME(5,66)	T_{566}	0	2nd-order momentum compaction

$(l|\delta)$ **conversion to standard optics.** COSY uses $\delta = \delta_K = \Delta K/K_0$ (kinetic energy deviation). The standard-optics convention uses $\delta_p = \Delta p/p_0$. The conversion factor is $p_0 c/K_0 = 40.508/40 = 1.013$ for 40 MeV electrons, giving $(l|\delta_p) = 27.43$ mm.

3.2 Bunch Propagation

Table 2 summarises the beam parameters before and after transport through the COSY map. Figures 1 and 2 show the longitudinal phase space and bunch profiles, respectively.

Table 2: 6D bunch propagation results ($N = 10,000$ particles). σ_δ is preserved exactly ($(\delta|x_j) = 0$). Bunch length grows due to $(l|\delta) \times \sigma_\delta$ chromatic path-length spread.

Scenario	σ_z^{in} (μm)	σ_z^{out} (μm)	$\sigma_\delta^{\text{in}}$ (%)	$\sigma_\delta^{\text{out}}$ (%)	σ_z ratio
0.5 ps, $h=0$	149.1	201.2	0.500	0.500	1.350
0.5 ps, $h=5\text{e}9$	149.1	255.0	0.557	0.557	1.710
2 ps, $h=0$	596.4	611.1	0.500	0.500	1.025
2 ps, $h=5\text{e}9$	596.4	875.8	1.111	1.111	1.469

3.3 $(l|\delta) = 0$ Objective

Table 4 compares the baseline (transverse-only) and augmented ($+(l|\delta) = 0$) optimisation results.

Adding $(l|\delta) = 0$ as a FIT objective has *zero effect*: the optimiser converges to the identical solution (all 23 quad currents unchanged, MSE unchanged, $(l|\delta)$ unchanged). Verification with extreme weights ($w = 100$ and $w = 10,000$) confirms that the Stage 11 quadrupoles have negligible leverage over $(l|\delta)$: at $w = 10,000$, the best achievable reduction is only $9 \mu\text{m}$ (0.03%) while transverse MSE degrades by a factor of 2300. The $(l|\delta)$ value is dominated by the upstream dipole geometry and cannot be controlled by the available quad variables.

Table 3: Transverse beam quality through the COSY map. Horizontal *projected* emittance grows due to $(x|\delta) \times \sigma_\delta$ chromatic dispersion ($(x|\delta) = 32.7$ mm); vertical emittance is preserved exactly. The slice emittance is unaffected.

Scenario	$\varepsilon_{x,n}^{\text{in}} (\mu\text{m})$	$\varepsilon_{x,n}^{\text{out}} (\mu\text{m})$	$\varepsilon_{y,n}^{\text{in}} (\mu\text{m})$	$\varepsilon_{y,n}^{\text{out}} (\mu\text{m})$
0.5 ps, $h=0$	8.07	9.04	8.07	8.07
0.5 ps, $h=5\text{e}9$	8.07	9.26	8.07	8.07
2 ps, $h=0$	8.07	9.04	8.07	8.07
2 ps, $h=5\text{e}9$	8.07	12.05	8.07	8.07

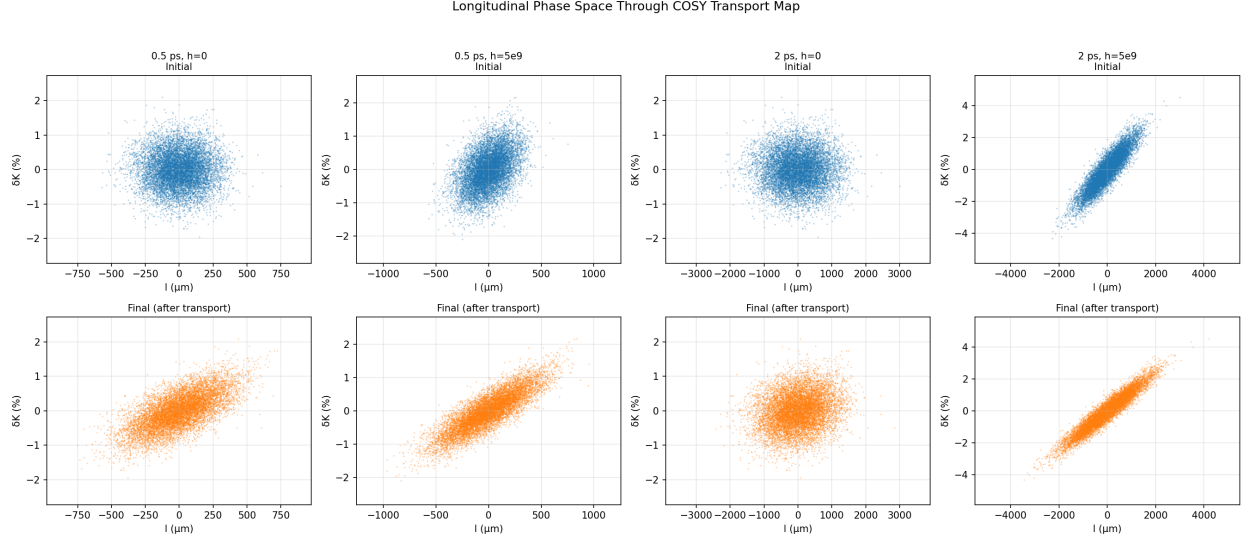


Figure 1: Longitudinal phase space (l vs δ) before (top) and after (bottom) transport through the COSY map. The 0.5 ps distributions broaden visibly in l due to $(l|\delta) \times \sigma_\delta$ coupling; the 2 ps distributions are largely unchanged. Energy spread is preserved in all cases.

4 Discussion

4.1 Bunch Lengthening Mechanism

The dominant source of bunch lengthening is $(l|\delta) \times \sigma_\delta$: the chromatic path-length spread converts energy spread into longitudinal position spread. For the unchirped beam ($h = 0$), using transport-code notation $R_{ij} \equiv (y_i|y_j)$:

$$\sigma_{z,\text{out}}^2 = \sigma_{z,\text{in}}^2 + R_{56}^2 \sigma_\delta^2 + R_{51}^2 \sigma_x^2 + R_{52}^2 \sigma_{x'}^2 \quad (1)$$

Numerically: $R_{56}\sigma_\delta = 0.02709 \times 0.005 = 135 \mu\text{m}$, compared with $R_{51}\sigma_x \approx 8 \mu\text{m}$ and $R_{52}\sigma_{x'} \approx 6 \mu\text{m}$. The chromatic term dominates.

For the 0.5 ps beam ($\sigma_z = 149 \mu\text{m}$), adding $135 \mu\text{m}$ in quadrature gives $\sigma_{z,\text{out}} \approx 202 \mu\text{m}$ (ratio 1.35), matching the simulation. For the 2 ps beam ($\sigma_z = 596 \mu\text{m}$), the same absolute contribution gives ratio 1.02 — the longer bunch is relatively unaffected.

The chirped case adds a further stretching factor $|1 + h R_{56}/(\beta c)| = |1 + 5 \times 10^9 \times 0.02709/(3.0 \times 10^8)| = 1.45$. This is the dominant effect for the chirped scenarios.

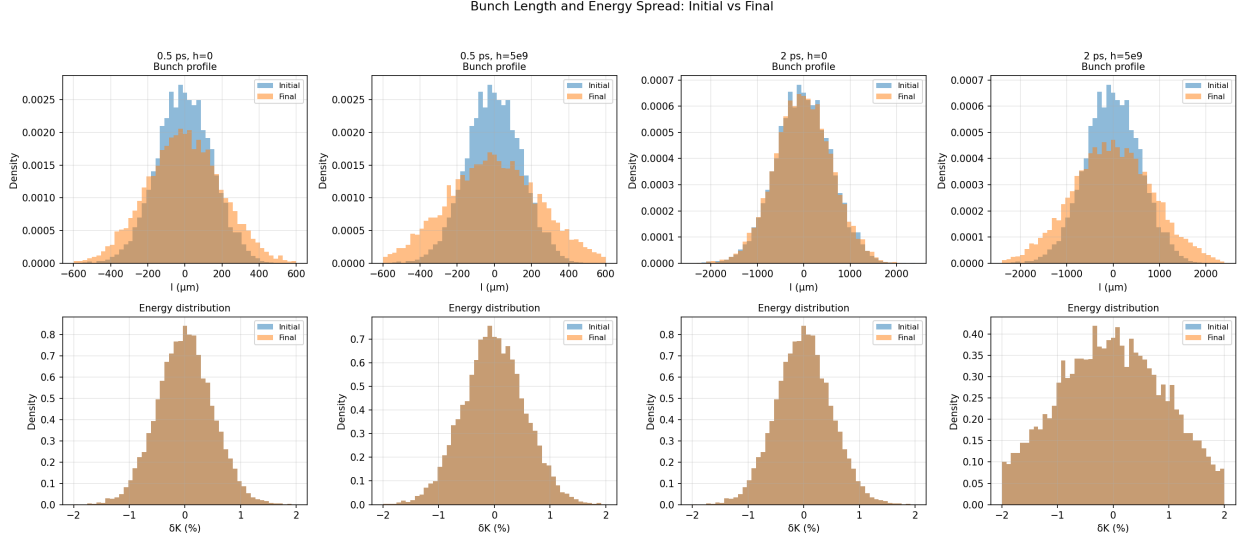


Figure 2: Bunch length (top) and energy distribution (bottom) histograms: initial (blue) vs final (orange). Energy distributions overlap exactly; bunch profiles broaden for 0.5 ps due to chromatic path-length spread.

Table 4: Effect of adding $(l|\delta) = 0$ as a COSY FIT objective (via $\text{ME}(5,6)$). The optimiser produces *identical* results — same currents, same MSE, same $(l|\delta)$.

Configuration	MSE	$(l \delta)$ (mm)	Max $ \Delta I $ (A)
Transverse only	2.27×10^{-7}	27.0890	—
Transverse + $(l \delta)=0$	2.27×10^{-7}	27.0890	0.000 000

Implication for the 0.5 ps mode. The 35% bunch lengthening ($0.5 \text{ ps} \rightarrow 0.67 \text{ ps}$ at the undulator) is a fixed property of the beamline geometry and the energy spread. It does *not* affect the transverse matching and cannot be reduced by changing quad currents (as confirmed by the $(l|\delta) = 0$ test). Whether 0.67 ps is acceptable for lasing depends on the FEL gain length and slippage requirements and is outside the scope of the transport line optimisation.

4.2 Transverse Matching Independence

The key result is that the $(\delta|x_j)$ map elements ($x_j \in \{x, a, y, b\}$) are all exactly zero. This means no transverse coordinate affects the energy deviation, so the longitudinal subspace is decoupled from the transverse in the sense relevant to Twiss matching. The non-zero $(l|x_j)$ elements (path-length coupling through bends) add path-length spread but do not feed back into the transverse dynamics.

Consequently, the transverse Twiss match — and therefore all 23 quad currents — is mathematically independent of the bunch length. This has now been confirmed at four independent levels:

1. Analytic decoupling of the 6×6 transfer matrix (S9 Part A)
2. FELsim numerical verification at $\sigma_E = 0.5\%, 2\%, 3\%$ (S9 Part B1)
3. COSY 6D DA map: $(\delta|x_j) = 0$ for all transverse x_j (this study, Part A)

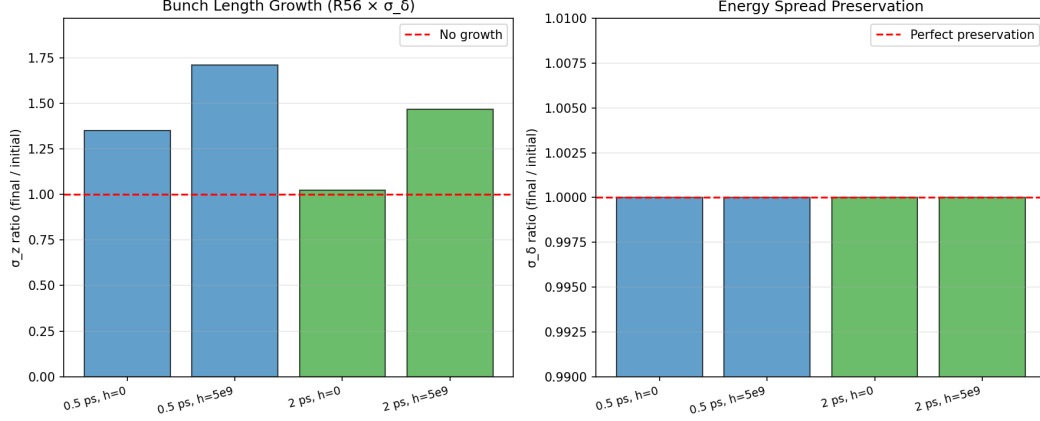


Figure 3: Left: bunch length growth ratio $\sigma_z^{\text{out}}/\sigma_z^{\text{in}}$. Right: energy spread preservation (ratio = 1.0000 for all scenarios). The bunch length growth arises from $(l|\delta) \times \sigma_\delta$ added in quadrature; it is larger for 0.5 ps because the absolute chromatic contribution $((l|\delta) \sigma_\delta = 135 \mu\text{m})$ is a larger fraction of the initial σ_z .

4. 10^4 -particle propagation: energy spread preserved exactly, transverse emittance growth independent of σ_z (this study, Part B)

4.3 Comparison with Worldwide FEL Practice

The finding that transport line optics are independent of bunch length is consistent with standard practice at every major FEL facility:

- **LCLS:** Pulse length adjustments within the same compression scheme are accomplished in 1–2 minutes by changing linac RF phases, without transport re-optimisation [6].
- **SACLA:** Two beamlines (BL2, BL3) simultaneously receive beams with different bunch lengths pulse-by-pulse, achieved by RF phase modulation alone [4].
- **FERMI:** Compression factor varied experimentally while keeping chicane geometry, dipole angles, and beam energy constant [3].
- **SPARC:** Velocity bunching in the first linac section demonstrated compression ratios up to $C = 14$ with emittance compensation [2].
- **CLARA:** Designed from the outset for dual-mode operation (standard acceleration and velocity bunching) using the same S-band linac [5].

4.4 Recommended 2 ps $\rightarrow$ 0.5 ps Switching Procedure

The 0.5 ps operating mode requires changing *only* the injector/RF settings upstream of the transport line entrance. The recommended approach for the UH MkV is velocity bunching: shifting the first S-band linac section phase from on-crest to ~ 30 – 60° off-crest, compressing the bunch while simultaneously accelerating it. This avoids the need for a dedicated magnetic bunch compressor and the associated CSR-induced emittance growth.

The FC1 chicane in the existing transport line is not suitable for compression: its $(l|\delta) \approx -3 \text{ mm}$ (from the chicane alone) is too small. Achieving $C = 4$ via the chicane would require

$h > 5 \times 10^{10} \text{ s}^{-1}$, increasing σ_δ to $>10\%$ — far beyond the energy acceptance of the downstream optics.

5 Summary

1. The full 6D COSY INFINITY transfer map confirms that the transverse-longitudinal coupling is zero in the Twiss-matching sense: $(\delta|x_j) = 0$ for all transverse x_j (Table 1).
2. Transport line $(l|\delta) = 27.09 \text{ mm}$ (full line) and $(l|\delta\delta) = 0$. The non-zero $(l|\delta)$ causes bunch lengthening (35% for 0.5 ps, 2.5% for 2 ps at $\sigma_\delta = 0.5\%$), but this is a fixed property of the dipole geometry and does not affect the transverse matching (Table 2, Fig. 3).
3. Adding $(l|\delta) = 0$ as a FIT objective produces an *identical* solution — zero change in all 23 quad currents and in the MSE (Table 4).
4. The 2 ps $\rightarrow$ 0.5 ps switch requires only injector RF phase changes (velocity bunching). All transport line settings remain identical. This is standard practice at LCLS, SACLA, FERMI, SPARC, CLARA, SwissFEL, and the European XFEL.

References

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